Week 5 — Completion/Stopping Output

Goal

Add clear completion or stopping rules and show them in the trail.

Required Outputs

☐ Completion/stopping rules implemented (tolerance/max iterations OR exact form reached)

☐ Trail states why the process stopped/completed

☐ At least 3 tests showing different completion outcomes

Evidence to Attach

☐ Screenshots: 2–3 runs showing different stopping/completion reasons

Reflection (Write short answers)

- Which stopping/completion rule is most reliable for your app?

Test Cases/Evidence to Attach

1.) **Method:** `Newton-Raphson`

Function: `x^2 - 2`

Initial Guess: `1.5`

Tolerance: `1e-6`

Max Iterations: `50`

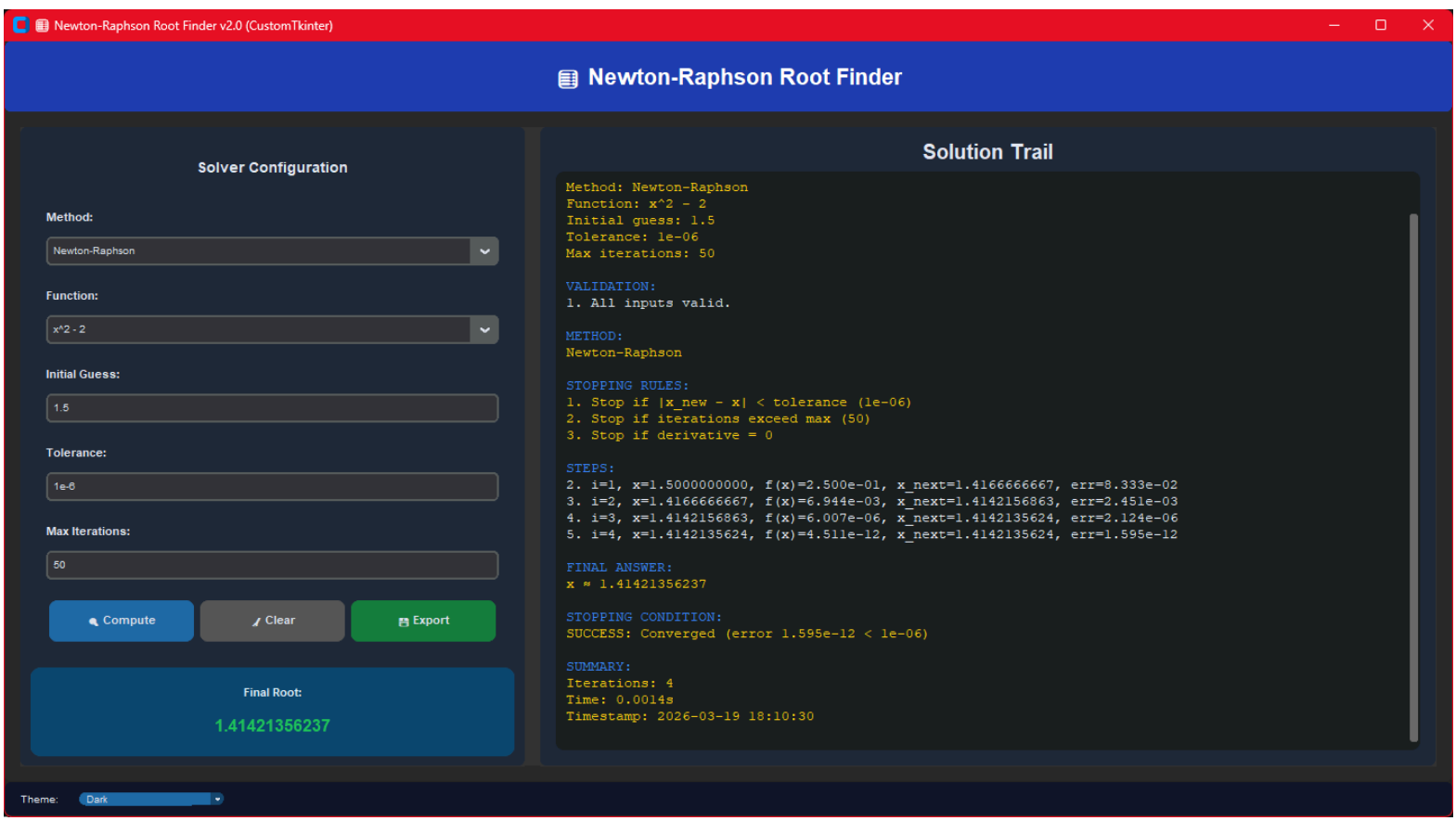
Expected Output

Final root ≈ `1.414213...`

Trail shows:

STOPPING CONDITION:

SUCCESS: Converged (error ... < tolerance)



Test Cases

2.) **Method:** `Newton-Raphson`

Function: `cos(x) - x`

Initial Guess: `10`

Tolerance: `1e-10`

Max Iterations: `3`

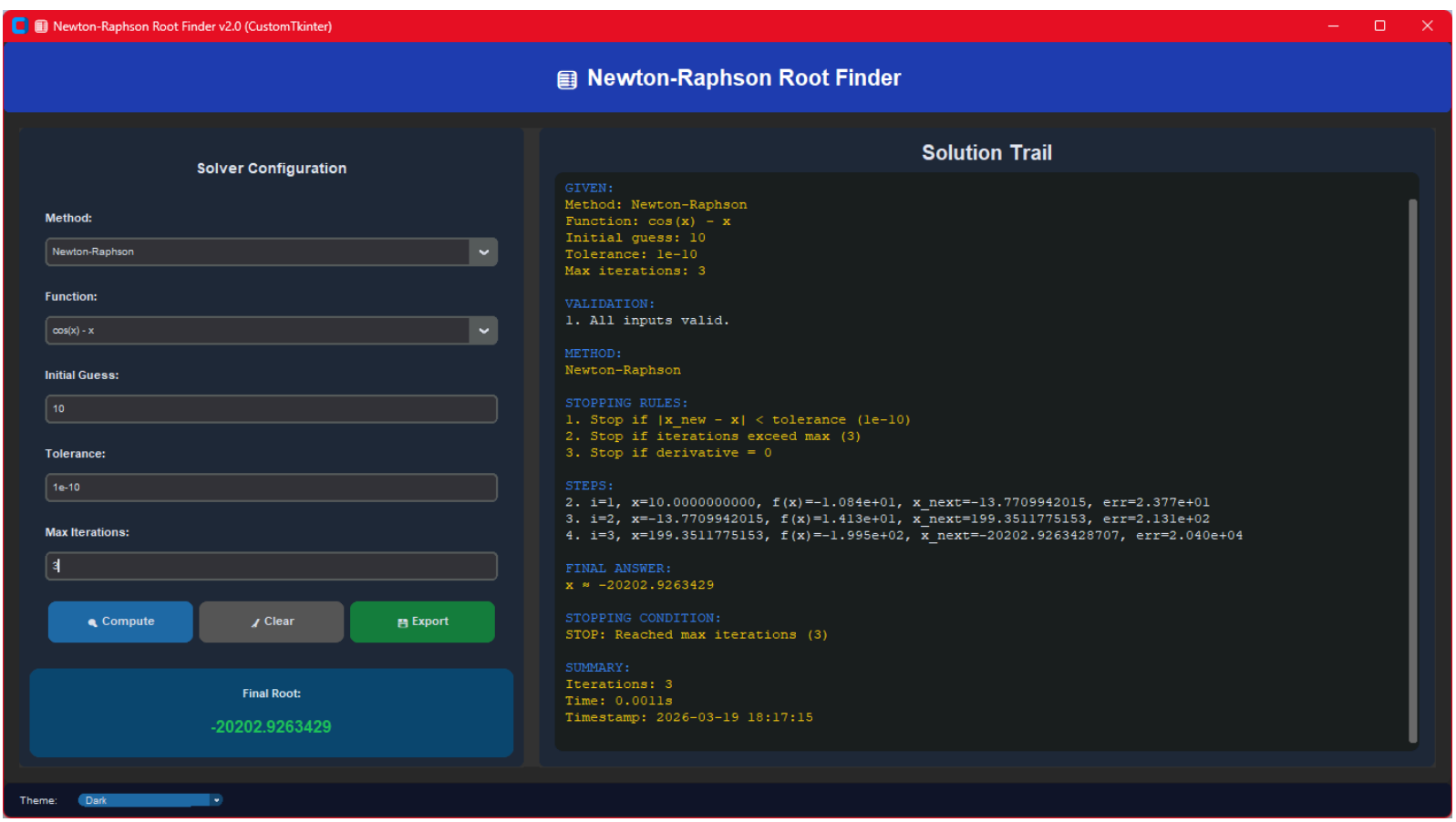
Expected Output

Final root ≈ `-20202.926...`

Trail shows:

STOPPING CONDITION:

STOP: Reached max iterations (3)



Test Cases

3.) **Method:** `Secant`

Function: `x^2 - 2`

Initial Guess: `1`

Second Guess: `1`

Tolerance: `1e-10`

Max Iterations: `50`

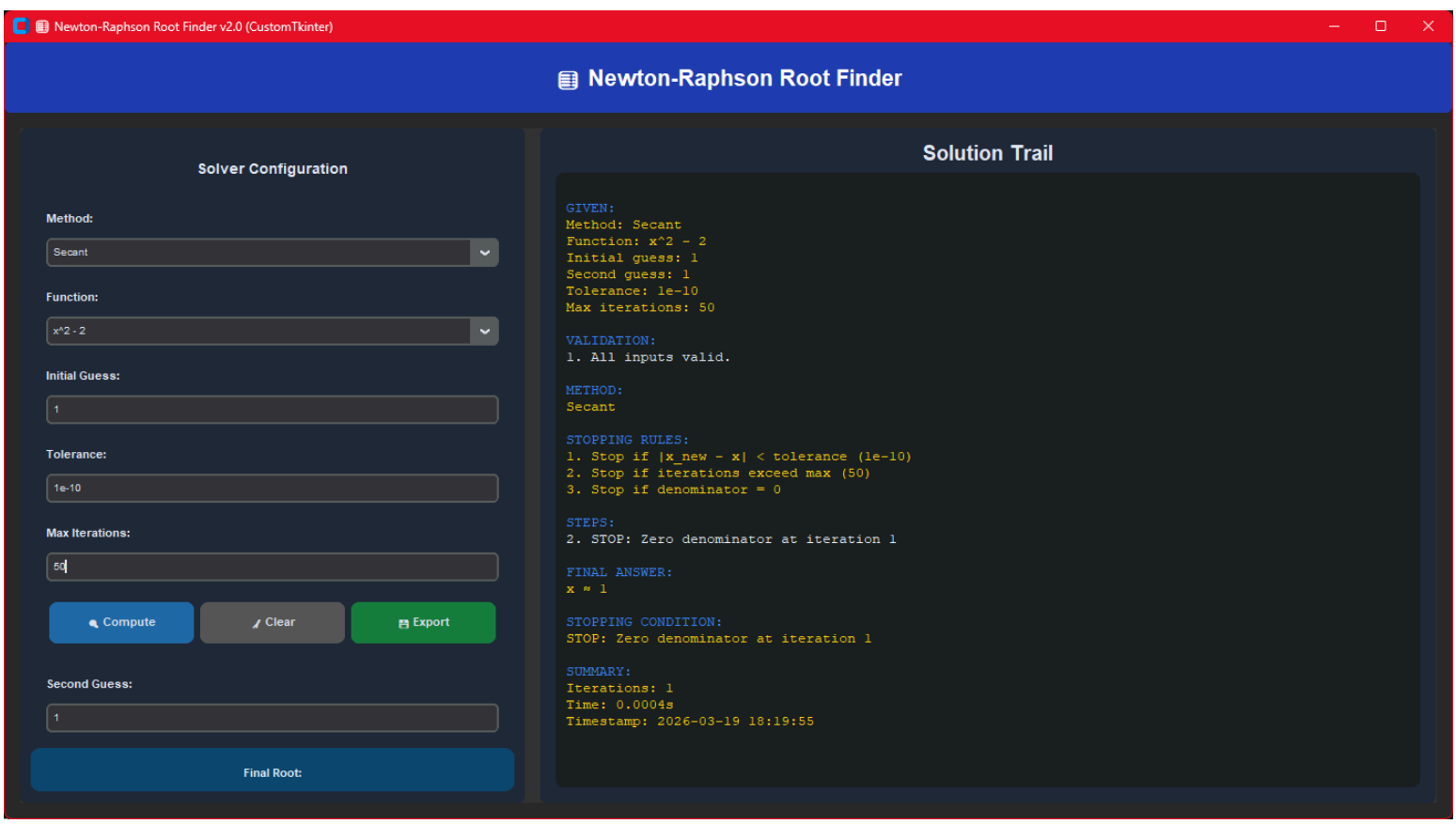
Expected Output

Final root ≈ `-20202.926...`

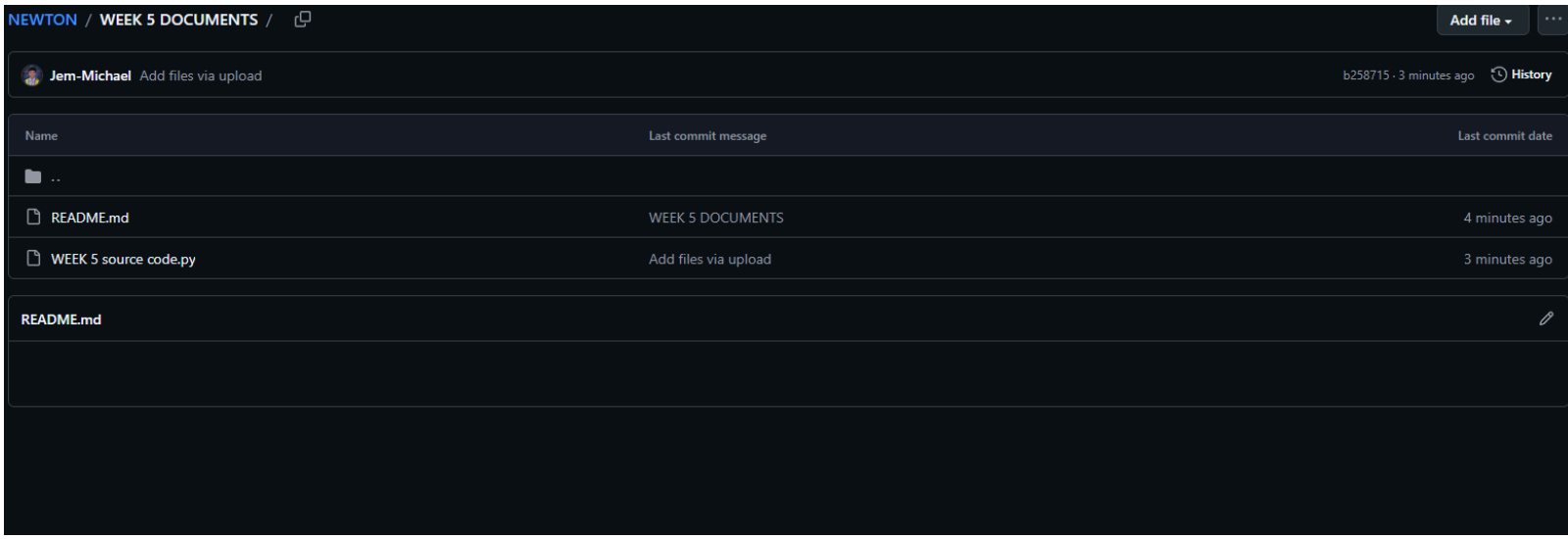
Trail shows:

STOPPING CONDITION:

STOP: Zero denominator at iteration 1



GITHUB REPO


NEWTON / WEEK 5 DOCUMENTS /
Add file
Jem-Michael Add files via upload
b258715 · 3 minutes ago
History
Name
Last commit message
Last commit date
..
README.md
WEEK 5 DOCUMENTS
4 minutes ago
WEEK 5 source code.py
Add files via upload
3 minutes ago
README.md

Reflection

Which stopping/completion rule is most reliable for your app?

The most reliable stopping or completion rule is the tolerance-based condition, because it directly measures how close the current answer is to the true root. If the difference between successive values is already very small, it means the solution is already accurate enough. Compared to just using the maximum number of iterations, tolerance is better because it does not stop too early or too late, it stops exactly when the desired accuracy is reached. The maximum iteration rule is still important as a backup, especially when the method does not converge, but it is not as reliable for accuracy. Other conditions like zero derivative or zero denominator are only for special cases or errors. With that, the tolerance condition is the most dependable because it ensures both accuracy and efficiency in finding the root.

Weekly Sign-off

Group Leader Name/Signature: CHARLES RUSSEL O. SAN JUAN Date: 3/19/2026

Member Name/Signature: CHRISTINA GOMBA Date: 3/19/2026

Member Name/Signature: JOHN MICHAEL TULALIAN Date:3/19/2026